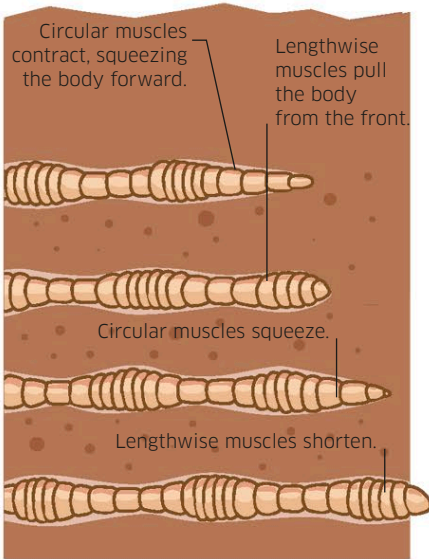


Burrowing

Life underground comes with special challenges. Burrowers need the strength to dig through soil to create a passage, and the ability to crawl through small openings. Moles use their feet like shovels to claw back the soil, but earthworms bulldoze their way through with their bodies.

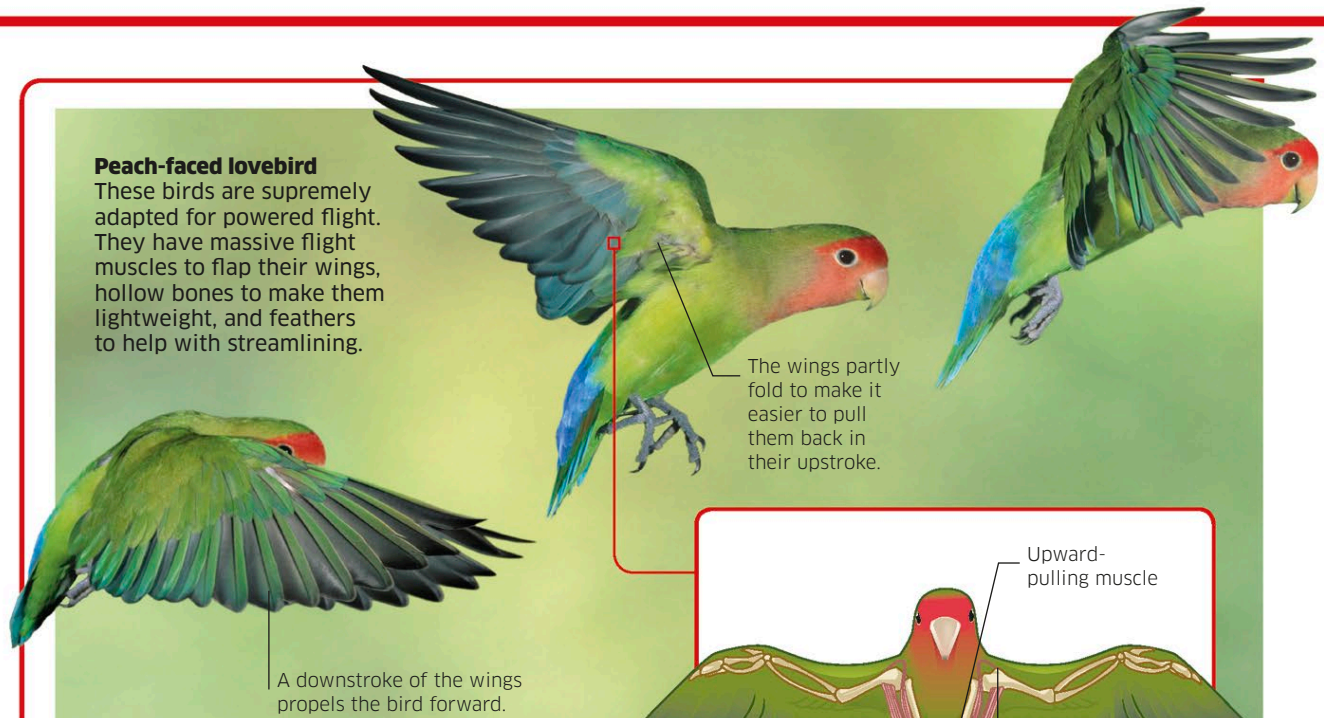


Earthworm

An earthworm has two sets of muscles. One set encircles the body and squeezes to push it forward, like toothpaste from a tube. The other pulls the body forward.

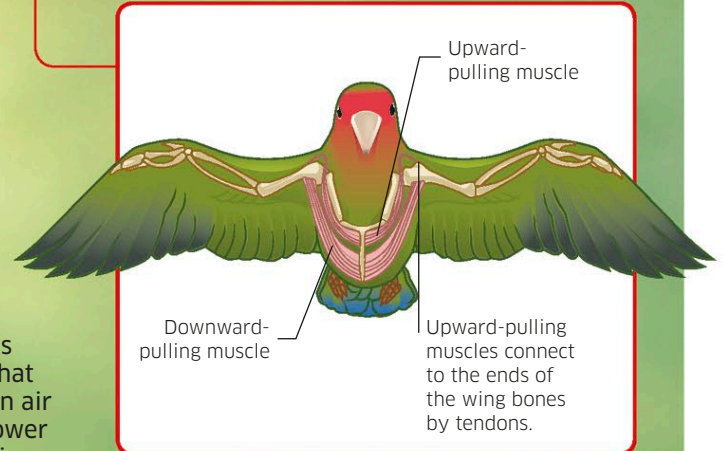
Peach-faced lovebird

These birds are supremely adapted for powered flight. They have massive flight muscles to flap their wings, hollow bones to make them lightweight, and feathers to help with streamlining.



Flying

The wings of flying animals are airfoils. This means they are slightly curved on top, so that air flowing over the wing travels faster than air passing beneath. Faster-moving air has a lower pressure, so air pressure underneath the wing is higher. It is this that causes the lift force and keeps the animal in the sky. Some airborne animals, such as flying squirrels, can only glide, but three groups flap their wings in powered flight: insects, bats, and birds.



The flying system

The largest chest muscles lower the wings by pulling down on the bones in the forearm. Other chest muscles connect on top of the forearm bones and pull the wings back up.

Swimming

Water is thicker than air, so it exerts a bigger force called drag against any animal that moves through it. Swimming animals reduce drag by being streamlined. Even though marine animals, such as fish and dolphins are only distantly related, they both have similar body shapes, to better propel themselves through the water.

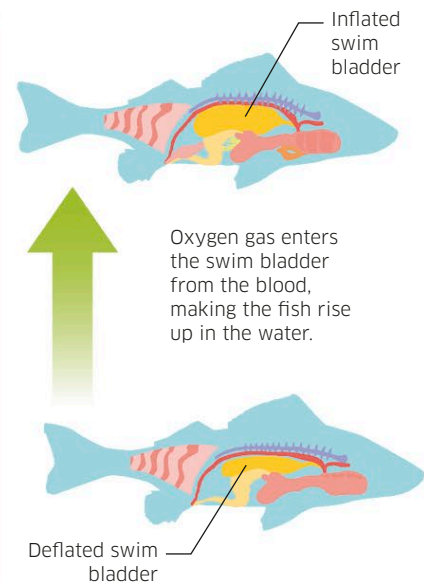
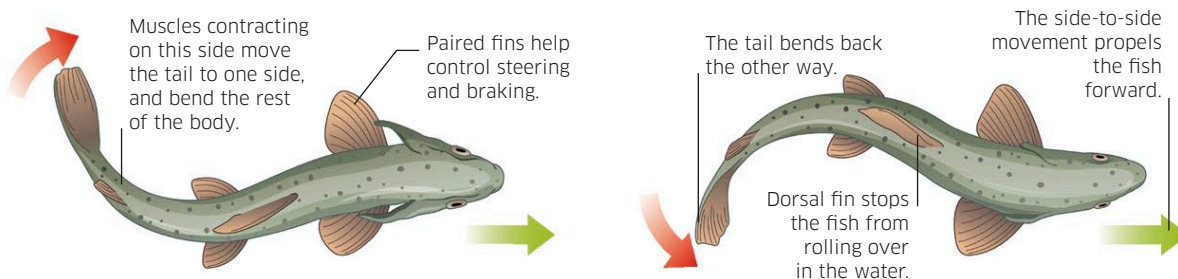
Swimming fish

Fish have blocks of muscle in the sides of their body. These contract to bend the body in an "S" shape, sweeping the tail from side to side and propelling the fish forward.



Sailfish

The enormous fin of a sailfish helps to steady its body—letting it get close to prey undetected. However, when the sail is lowered, it gives chase faster than any other fish in the ocean.



Controlling buoyancy

Fish are heavier than water, but most bony fish have a gas-filled chamber—the swim bladder—for staying buoyant when swimming. By controlling the volume of gas inside the swim bladder, fish can rise or sink through different water levels.